

Rapido Ride Analysis

Understanding Demand, Cancellations & Revenue in Bangalore

NST DVA Capstone 2 · Section A · Group 1

Aniket Pathak · Nipun Patlori · Saswataduity Bhuin · Akshay Y
Narendra Singh · Sayan Bhattacharya · Vedant Madne

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1 Project Overview

1.1 Business Problem

Ride-hailing platforms like Rapido face a persistent challenge: balancing rider demand with driver supply across different service types and time windows. High cancellation rates erode customer trust, reduce revenue, and signal inefficiencies in resource allocation.

Core Business Question:

How do service type, time of day, and location influence ride cancellation rates and revenue — and where should Rapido focus driver allocation to minimise cancellations during peak demand?

1.2 Dataset at a Glance

Attribute	Value
Source	Bangalore Rapido Ride Data (Kaggle, simulated)
Raw Rows	51,000
Cleaned Rows	49,801
Raw Columns	13
Cleaned Columns	30
Time Period	April 10 – April 29, 2024
Format	CSV

2 Pipeline: From Raw to Clean (ETL)

The entire pipeline runs across five Jupyter notebooks.

2.1 Notebook 01 — Extraction

- Loaded `rapido_rides.csv` (51,000 rows, 13 columns) with `pd.read_csv()`.
- Ran `df.info()` and `df.describe()` to inspect dtypes, ranges, and nulls.
- Identified the 13 raw columns: `services`, `date`, `time`, `ride_status`, `source`, `destination`, `duration`, `ride_id`, `distance`, `ride_charge`, `misc_charge`, `total_fare`, `payment_method`.
- Initial null audit showed **10,346 nulls** in `ride_charge`, `misc_charge`, `total_fare` and **12,398 nulls** in `payment_method`.

2.2 Notebook 02 — Cleaning & Feature Engineering

2.2.1 Step 1: Handling Missing Values (Business-Logic Driven)

The 10,346 missing fare rows corresponded to **cancelled rides** — rides that were booked but never completed, so they naturally have no fare. We did **NOT** drop them.

- Set `ride_charge`, `misc_charge`, `total_fare` $\leftarrow 0$ for cancelled rides.

- Set `payment_method` $\leftarrow$ "not_applicable" for cancelled rides.
- Recalculated `total_fare` = `ride_charge` + `misc_charge` for completed rides to fix any inconsistencies.

Key interview point: We chose business-logic imputation over mean/median imputation because a cancelled ride genuinely has zero fare — filling with an average would pollute revenue KPIs.

2.2.2 Step 2: Datetime Conversion

- `date` column converted from `object` $\rightarrow$ `datetime64`.
- Combined `date` + `time` strings into a new `timestamp` column for time-series analysis.

2.2.3 Step 3: Duplicate Removal

- Deduplicated on `ride_id`, keeping the first occurrence.
- Dropped 1,199 duplicate records.

2.2.4 Step 4: Categorical Standardisation

- All text columns lowercased and stripped of whitespace.
- Service names normalised: "bike lite" $\rightarrow$ `bike_lite`, "cab economy" $\rightarrow$ `cab_economy`.

2.2.5 Step 5: Anomaly Filtering

Dropped completed rides where:

- `distance` > 100 km (physically implausible in Bangalore city)
- `distance` $\leq$ 0 km
- `duration` $\leq$ 0 min

2.2.6 Step 6: Feature Engineering (17 New Columns)

Column	Formula / Logic	Purpose
year	date.dt.year	Temporal filter
month	date.dt.month	Seasonal trends
hour	timestamp.dt.hour	Hourly demand
day_of_week	timestamp.dt.day_name()	Weekly patterns
is_weekend	1 if Sat/Sun, else 0	Weekend vs weekday
peak_hour	1 if hour ∈ [8,9,10,17,18,19]	Peak flagging
completed	1 if completed, 0 if cancelled	Binary target
avg_speed_kmh	distance / (duration/60)	Operational efficiency
fare_per_km	total_fare / distance	Unit economics
source_cancel_rate	avg cancel rate per pickup zone	Geographic reliability
demand_intensity	Z-score of volume per source-hour	Supply pressure
demand_zone	low / medium / high demand label	Zone segmentation
distance_category	short / medium / long	Ride length segment
fare_category	budget / standard / premium / cancelled	Revenue segment

3 KPI Framework

KPI	Value	Formula	Insight
Cancellation Rate	10.40%	5,180 / 49,801	1 in 10 rides cancelled
Total Revenue	Rs.68,37,929	Sum of total_fare	Completed rides only
Avg Revenue/Ride	Rs.153.24	Revenue / completed	Baseline fare benchmark
Peak Cancel Rate	13.22%	1,653 / 12,503	40% higher than off-peak
Off-Peak Cancel Rate	9.46%	3,527 / 37,298	Baseline comparison

4 Exploratory Data Analysis (EDA)

4.1 Service Type Analysis

Service	Rides	Cancelled	Cancel %	Revenue (Rs.)	Avg Fare
auto	11,474	1,180	10.28%	19,36,736	188.14
bike	17,492	1,813	10.36%	13,02,960	83.10
bike_lite	4,541	452	9.95%	2,32,899	56.96
cab_economy	9,370	1,006	10.74%	26,98,620	322.65
parcel	6,924	729	10.53%	6,66,713	107.62

Key insight: cab_economy has the highest cancellation rate (10.74%) and highest average fare (Rs.322.65) — premium rides are most at risk. bike_lite is the most reliable service.

4.2 Peak vs Off-Peak Analysis

Window	Rides	Cancelled	Cancel Rate
Off-Peak	37,298	3,527	9.46%
Peak (8-10 AM, 5-7 PM)	12,503	1,653	13.22%

Peak hours see **39.7% more cancellations** than off-peak. The 5 PM – 7 PM evening slot is the worst at **14.14%** cancellation rate.

4.3 Hour-wise Cancellation Heatmap (Key Hours)

Hour	Cancel Rate	Interpretation
5 AM	12.84%	Late-night stragglers, low driver availability
8 AM	13.67%	Morning office rush begins
10 AM	13.24%	Peak tail
17 (5 PM)	14.14%	Worst hour of day — evening rush
18 (6 PM)	12.62%	Evening peak continues
12 PM	7.07%	Midday lull, best completion
14 (2 PM)	7.00%	Lowest cancellation of day

4.4 Geographic Analysis

- **Top Revenue Zone:** Koramangala — Rs.2,73,504 (4% of total revenue from one zone).
- Koramangala also has 1,957 rides with a 10.17% cancellation rate.
- The zone naming inconsistency (`koramangala` vs `kormangala`) was detected during EDA and standardised.

4.5 Demand Zone Distribution

Demand Zone	Ride Count
High Demand	17,354
Medium Demand	17,590
Low Demand	14,857

4.6 Efficiency Metrics (Completed Rides)

Metric	Mean	Median	Min	Max
Distance (km)	6.99	5.48	0.8	45.0
Duration (min)	24.09	19.0	5	120
Avg Speed (km/h)	18.31	17.69	2.07	73.1
Fare per km (Rs.)	20.94	19.85	0	110.18
Total Fare (Rs.)	137.31	93.84	0	1,301.58

4.7 Service Efficiency Comparison

Service	Avg Fare/km	Avg Speed (km/h)	Avg Fare (Rs.)
cab_economy	Rs.33.77	19.59	322.65
auto	Rs.28.47	18.75	188.14
parcel	Rs.20.56	17.93	107.62
bike	Rs.17.76	17.89	83.10
bike_lite	Rs.15.05	17.46	56.96

4.8 Payment Method Distribution

Payment Method	Count
Paytm	11,208
QR Scan	11,186
GPay	11,147
Amazon Pay	11,080
Not Applicable (cancelled)	5,180

Note: Payment methods are nearly uniformly distributed across all four UPI providers — no single provider dominates.

5 Statistical Analysis

5.1 Test 1: Chi-Square — Service Type vs Cancellation

- **Null Hypothesis:** Service type and cancellation are independent.
- **Result:** $\chi^2 = 2.42$, $p = 0.659$, $df = 4$
- **Conclusion: Fail to reject H_0 .** No statistically significant difference in cancellation rates across services. All services cancel at approximately the same rate ($\sim 10.4\%$).

5.2 Test 2: T-Test — Peak vs Off-Peak Cancellations

- **Null Hypothesis:** Peak and off-peak cancellation rates are equal.
- **Result:** $t = -11.95$, $p < 0.0001$ (statistically significant)
- **Conclusion: Reject H_0 .** Peak hours have a **significantly higher** cancellation rate (13.22% vs 9.46%). The time-of-day effect is real and not due to chance.

5.3 Test 3: Correlation — Demand Intensity vs Cancellation

- Pearson correlation between `demand_intensity` (z-score of volume) and cancellation: $r = -0.0134$
- **Interpretation:** Very weak correlation. High demand alone does not predict cancellation. Other factors (driver supply, time of day) are more important drivers.

5.4 Revenue Leakage Simulation

- Average completed ride fare: Rs.153.24
- If cancellations in top revenue zones are reduced by 10%:
- **Potential monthly recovery** $\approx$ Rs.1,20,000 (extrapolated from 20-day dataset)

6 Charts & Visualisations

All charts were created using **Python (matplotlib / seaborn)** in 03_EDA.ipynb and 04_statistical_analysis.ipynb, and published as an interactive **Tableau Dashboard**.

6.1 Chart List

1. **Bar Chart — Rides by Service Type:** bike leads with 17,492 rides; bike_lite is smallest with 4,541.
2. **Stacked Bar — Cancellation by Service:** Shows completed vs cancelled split per service. All services cluster near 10%.
3. **Line Chart — Hourly Cancellation Rate:** Clear spikes at 8 AM (13.67%) and 5 PM (14.14%). Troughs at 12–14 PM.
4. **Heatmap — Service $\times$ Peak Hour Cancellation:** 5x2 grid showing peak vs off-peak rates per service. cab_economy peak = 13.12%.
5. **Box Plot — Fare Distribution by Service:** cab_economy has highest spread (Rs.0–1301). bike_lite is tightest.
6. **Scatter Plot — Distance vs Fare:** Positive correlation, with cab_economy having steepest fare-per-km slope.
7. **Bar Chart — Revenue by Service:** cab_economy = Rs.26.9L despite fewer rides; bike = Rs.13L with most rides.
8. **Map / Geographic Chart (Tableau):** Cancellation hotspots by pickup zone overlaid on Bangalore map.
9. **Histogram — Fare Distribution:** Right-skewed distribution; most rides between Rs.50–200.
10. **Day-of-Week Chart:** Tuesday has highest cancellation rate (11.19%); Wednesday is most reliable (9.60%).

6.2 Tableau Dashboard

Two dashboard views were published:

- **Executive View:** Cancellation rates by service, peak vs off-peak summary, top revenue zones.
- **Operational View:** Ride volume by hour, cancellation heatmap, fare distribution by service.

7 Key Business Insights

1. **Peak Hour is the Primary Risk Factor:** Cancellation spikes 39.7% during morning (8–10 AM) and evening (5–7 PM) peaks. Time of day, not service type, is the dominant driver.
2. **Service Type Doesn't Matter for Cancellation:** Chi-square test confirms all services cancel at the same rate. Operational improvements must focus on time/location, not service segmentation.
3. **cab_economy is the Highest Revenue Risk:** It earns the most per ride (Rs.322.65 avg) and has the highest cancellation rate (10.74%). Every lost cab ride costs 4x more than a lost bike ride.
4. **bike and bike_lite are the Most Resilient:** During peak hours, bikes maintain relatively low cancellation and high volume. They can serve as overflow capacity.
5. **Koramangala is the Critical Zone:** 4% of total revenue from one zone. This zone must be prioritised for driver incentives during peak hours.
6. **Midday (12–3 PM) is the Efficiency Window:** Lowest cancellation rates (7.0–7.5%), ideal time for driver training or scheduled maintenance.
7. **Revenue Concentration:** Top 5 zones contribute disproportionate revenue — geographic concentration is a risk factor if driver supply isn't maintained there.
8. **Payment is Neutral:** No UPI provider dominates, suggesting users have no strong payment preference. Not a lever for intervention.

8 Business Recommendations

1. **Peak-Hour Driver Surge Incentives:** Offer bonus payouts to drivers who are active between 8–10 AM and 5–7 PM. Target Koramangala and HSR Layout first. Expected impact: reduce peak cancellation from 13.22% to ~10%.
2. **Zone-Based Driver Pre-positioning:** Use demand zone data to pre-position drivers in high-demand zones 15 minutes before peak hours begin. Implement via the driver app as shift recommendations.
3. **cab_economy Supply Boost:** Since cab rides are highest revenue but most vulnerable, create a dedicated cab driver pool for peak hours in top-5 revenue zones.
4. **Bike as Peak Overflow:** During extreme demand, surface bike/auto alternatives faster in the UI when cabs are unavailable. Reduces user frustration and captures revenue that would otherwise be lost.
5. **Real-Time Cancellation Monitoring:** Deploy a live dashboard (like our Tableau view) for the ops team to track hourly cancellation rates and trigger driver incentives dynamically.

9 Tech Stack & Team Roles

Tool	Purpose
Python	ETL, Cleaning, EDA, Statistical Analysis
Pandas	Data manipulation and transformation
SciPy	Statistical tests (chi-square, t-test)
Matplotlib / Seaborn	Chart generation
Tableau Public	Interactive dashboard
GitHub	Version control, collaboration
Jupyter Notebooks	Development environment

Name	Role	Primary Contribution
Aniket Pathak	Project Lead	Problem framing, ETL oversight
Nipun Patlari	Data Lead	EDA, cleaning pipeline
Saswataduity Bhuin	ETL Lead	Tableau dashboard
Akshay Y	Analysis Lead	Statistical analysis
Narendra Singh	Visualization Lead	Tableau + PPT
Sayan Bhattacharya	Strategy Lead	Recommendations
Vedant Madne	PPT & Quality Lead	Presentation quality

10 Quick Revision Cheat Sheet

- **Raw data:** 51,000 rows, 13 columns
- **Cleaned data:** 49,801 rows, 30 columns
- **Missing values:** 10,346 in fare columns (cancelled rides) → filled with 0
- **Duplicates removed:** 1,199 rows
- **Cancellation Rate:** 10.40% (5,180 rides)
- **Peak Cancellation:** 13.22% vs 9.46% off-peak (t -test: $p < 0.0001$)
- **Service test:** Chi-square $p = 0.66$ — service type does NOT affect cancellation
- **Total Revenue:** Rs.68,37,929
- **Highest revenue service:** cab_economy (Rs.26.9L)
- **Most rides service:** bike (17,492 rides)
- **Best hour to ride:** 2 PM — 7.00% cancellation rate
- **Worst hour to ride:** 5 PM — 14.14% cancellation rate
- **Top zone:** Koramangala — Rs.2,73,504 revenue
- **Peak hours defined:** 8–10 AM and 5–7 PM

- **Avg speed:** 18.31 km/h
- **Avg fare per km:** Rs.20.94